As in Section 27.6, we introduce a pair of gauge-invariant left-chiral external superfields, now called Y and T, and replace the Lagrangian density with

$$\mathscr{L}^{\sharp} = \left[\Phi^{\dagger}\, e^{-V}\, \Phi\right]_{D} + 2\,\mathrm{Re}\left[Y f(\Phi)\right]_{\mathscr{F}} + \mathrm{Re}\left[\frac{T}{8\pi i}\sum_{A\alpha\beta}\epsilon_{\alpha\beta}\,W_{A\alpha L}\,W_{A\beta L}\right]_{\mathscr{F}} . \tag{29.3.3}$$

This becomes the same as (29.3.1) when we set the spinor and auxiliary components of Y and T equal to zero, and take their scalar components as $y = 1$ and $t = \tau$, respectively. Non-perturbative effects will in general invalidate both of the two symmetries on which the analysis of Section 27.6 was based. The translation operation, which in our present notation is $T \to T + \xi$ with real ξ, is not a symmetry because $\sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma}$ can have a non-vanishing integral over spacetime. The original R invariance (with T and Y having R values 0 and +2) is not a symmetry because the anomaly discussed in Chapter 22 gives the R-current a non-vanishing divergence. With θ_L and θ_R having $R = +1$ and $R = -1$, respectively, and V_A and Φ_n R-neutral, the fermion fields λ_{AL} and ψ_{nL} have $R = +1$ and $R = -1$, respectively, so Eq. (22.2.26) here gives

$$\partial_\mu J_R^\mu = -\frac{1}{32\pi^2}\Big(C_1 - C_2\Big)\sum_A \epsilon_{\mu\nu\rho\sigma} f_A^{\mu\nu} f_A^{\rho\sigma} , \tag{29.3.4}$$

where C_1 and C_2 are the constants defined in Eqs. (17.5.33) and (17.5.34):

$$\sum_{CD} C_{ACD} C_{BCD} = C_1\, \delta_{AB} , \qquad \mathrm{Tr}\left\{t_A t_B\right\} = C_2\, \delta_{AB} , \tag{29.3.5}$$

with the trace taken over all species of left-chiral superfield.* For instance, in the generalized supersymmetric version of quantum chromodynamics studied in Reference 3, with gauge group $SU(N_c)$ and N_f pairs of left-chiral quark superfields Q_n and $\overline{Q}_n$ in the defining representation and its complex conjugate, these constants have values given by Eq. (17.5.35) (with $n_f = 2N_f$) as

$$C_1 = N_c , \qquad C_2 = N_f .$$

Although T translation and R invariance are invalidated by non-perturbative effects, there is a remaining symmetry which is almost as

* A factor 32 instead of 16 appears in the denominator in Eq. (29.3.4) because gauginos do not have distinct antiparticles, and we are now counting antiparticles separately from particles in taking the trace in Eq. (29.3.5). Also, we are now adopting the convention, described at the end of Section 27.3, of including a gauge coupling factor in the gauge fields and not in structure constants and the matrix generators t_A. The gauge generators are thus normalized so that for t_A, t_B, and t_C in the standard $SU(2)$ subalgebra of the gauge algebra, the structure constant is $C_{ABC} = \epsilon_{ABC}$.